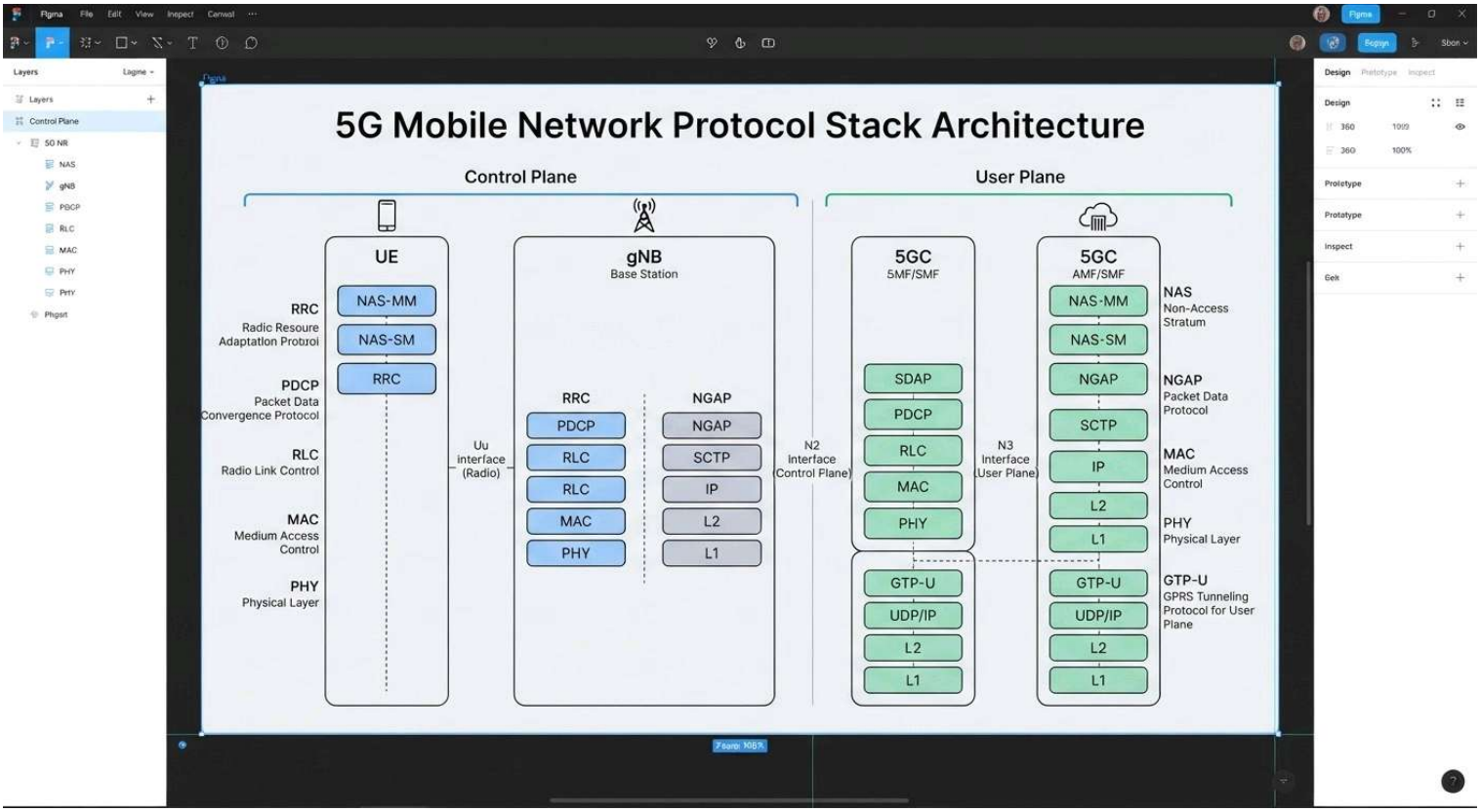

5G Mobile Network Protocol Stack Architecture
Control Plane
User Plane
UE
gNB
Base Station
5GC
5MF/SMF
5GC
AMF/SMF
NAS-MM
NAS-SM
RRC
PDCP
RLC
MAC
PHY
SDAP
NGAP
SCTP
IP
L2
L1
GTP-U
UDP/IP
RRC
Radio Resoure Adaptation Protocol
PDCP
Packet Data Convergence Protocol
RLC
Radio Link Control
MAC
Medium Access Control
PHY
Physical Layer
Uu interface (Radio)
N2 Interface (Control Plane)
N3 Interface (User Plane)
NAS
Non-Access Stratum
NGAP
Packet Data Protocol
MAC
Medium Access Control
PHY
Physical Layer
GTP-U
GPRS Tunneling Protocol for User Plane